

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
	Shut off cooling water
	Shut off fuel oil
	Shut off lubricating oil
	Lock turbocharger rotors

[illegible]



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description

1. Turn the engine in AHEAD direction to bring the piston of cylinder No. 1 to TDC.

Check with the pin gauge.

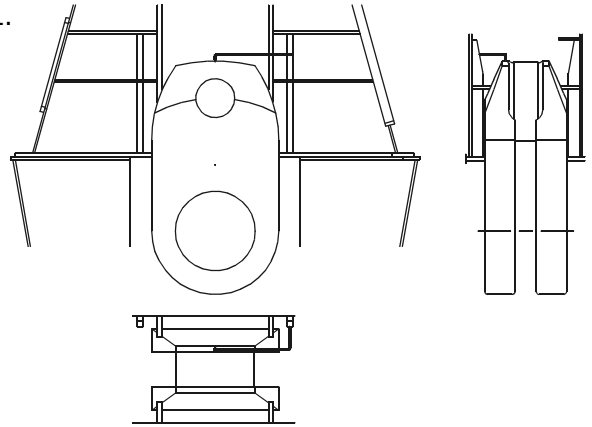
2. Remove the plug screw and gasket from the side of the distributor housing, and insert check pin **K** in slot **S**.

If pin **K** is not available, use the cylindrical end of a drill of a suitable size.

3. If the check pin cannot enter the slot, i.e. if the slot is not aligned with the hole in the distributor housing, the air distributor must be adjusted.

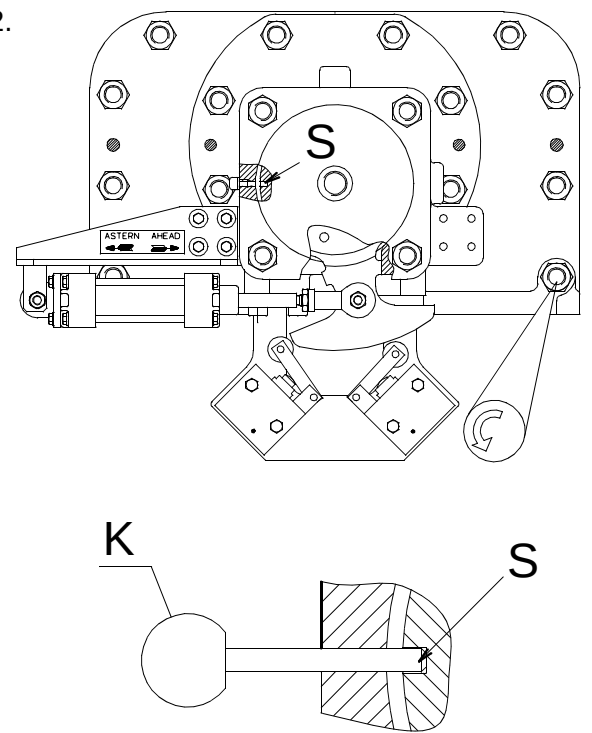
See Procedure 907-1.3.

1.

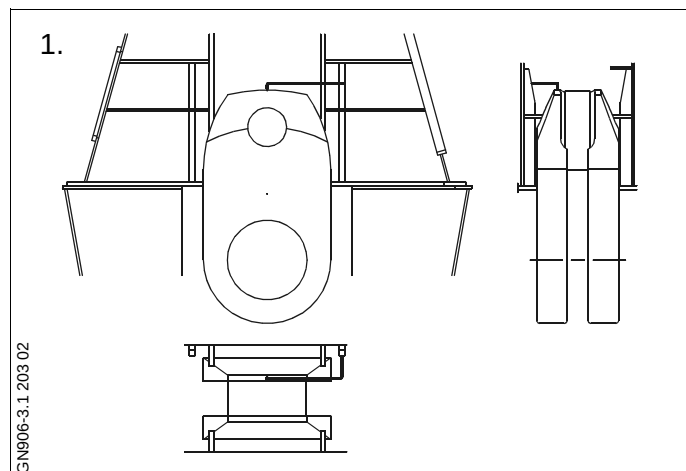


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2.



EN907-1.3 204 02

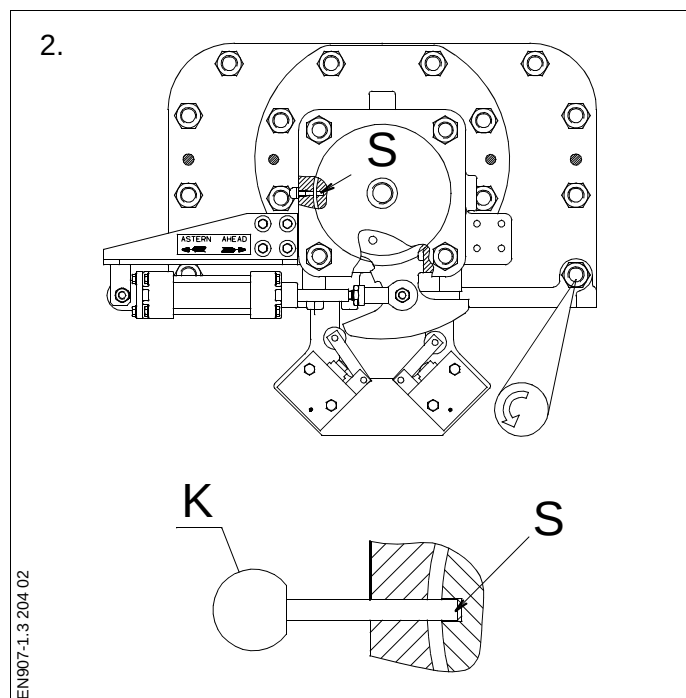


1. Turn the engine in AHEAD direction to bring the piston of cylinder 1 to TDC.

Check with the pin gauge.

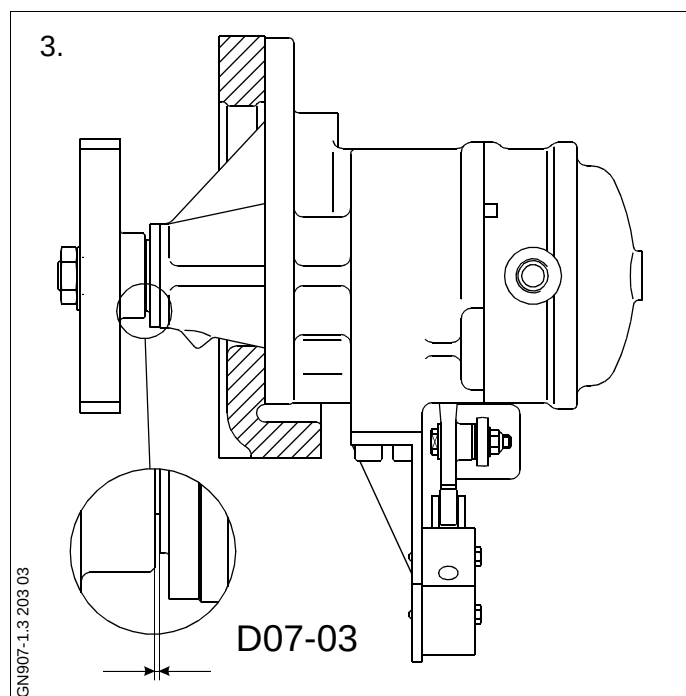
2. Dismount the end cover of the camshaft housing, along with the starting air distributor, from the engine.

Remove the plug screw and gasket from the side of the distributor housing, and insert check pin **K** in slot **S**.



Turn the shaft together with the distributor disc until check pin **K** enters slot **S** in the disc and keep it there.

3. Check and, if necessary, adjust clearance D07-03 between the gear wheel and the bushing.

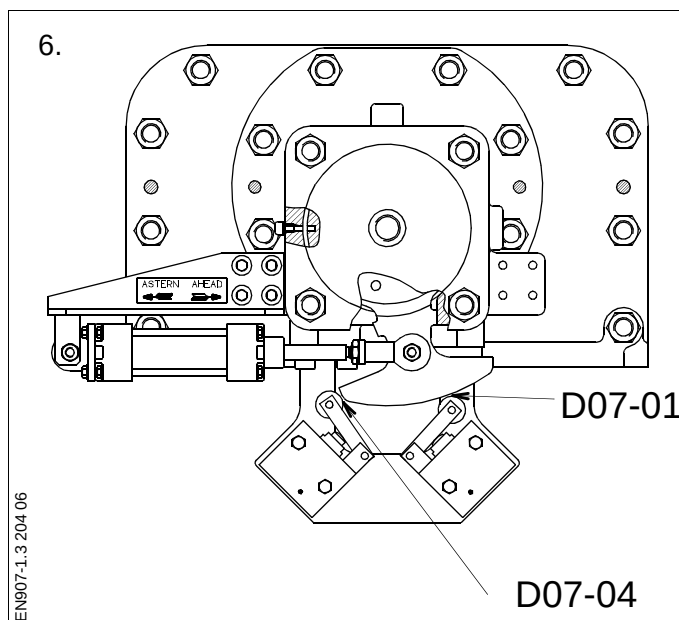
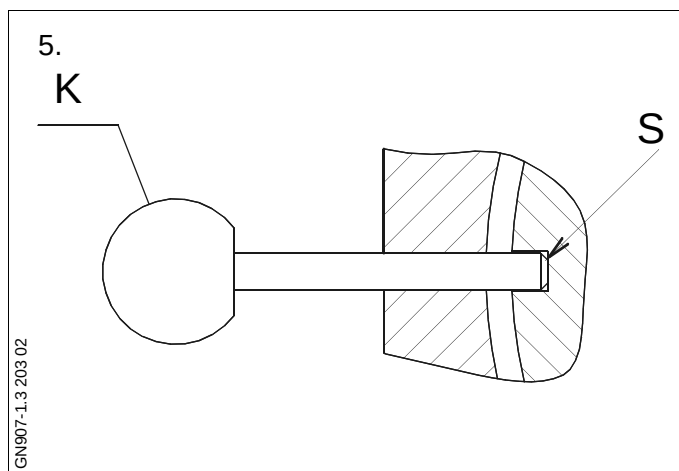
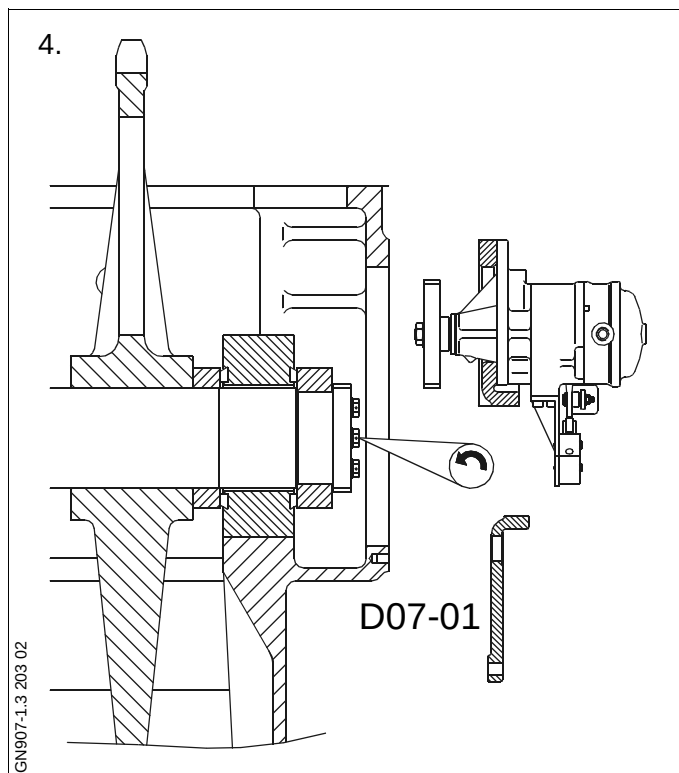


4. Remove the lower half of the gearwheel drive cover. Mount the upper half of the gearwheel drive cover with the complete starting air distributor after loosening the gear wheel on the camshaft, and then engage the two gear wheels by turning the wheel on the camshaft.

Tighten and lock the gear wheel screws on the camshaft, see *Data, D07-01*.

Mount the lower half of the gearwheel drive cover.

5. Remove check pin **K** from the distributor housing and mount the gasket and plug screw.
6. Check the clearance between the roller and the two interlocking valves with the roller in its innermost and outermost positions, respectively (ASTERN and AHEAD).



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D07-09	Piston tightening torque	180	Nm
D07-10	Valve tightening torque	175±20	Nm
D07-11	Valve tightening angle	50/30	°
D07-12	Starting air valve	24	kg

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Plate	Item No.	Description
P90151	095	Grinding ring for starting valve
P90151	105	Grinding handle for starting valve
P91359		Torque Spanners

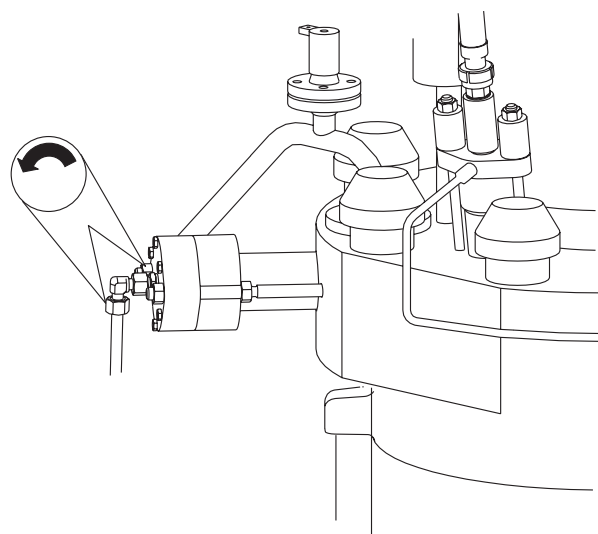
1. Shut off the starting air and control air inlet.

Dismount the control air pipes.

Unscrew the fixing nuts of the starting valve flange.

2. Pull the starting air valve out of the cylinder cover and remove it from the engine.

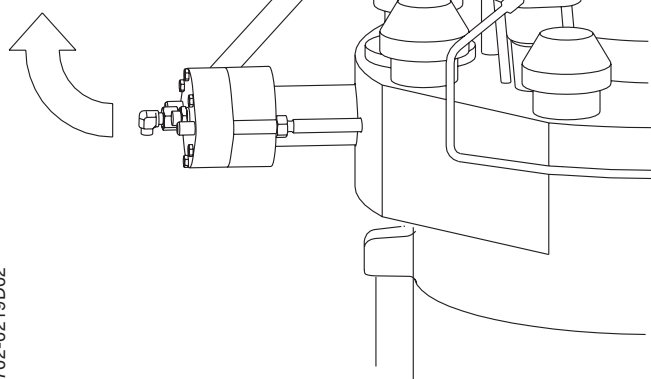
1.



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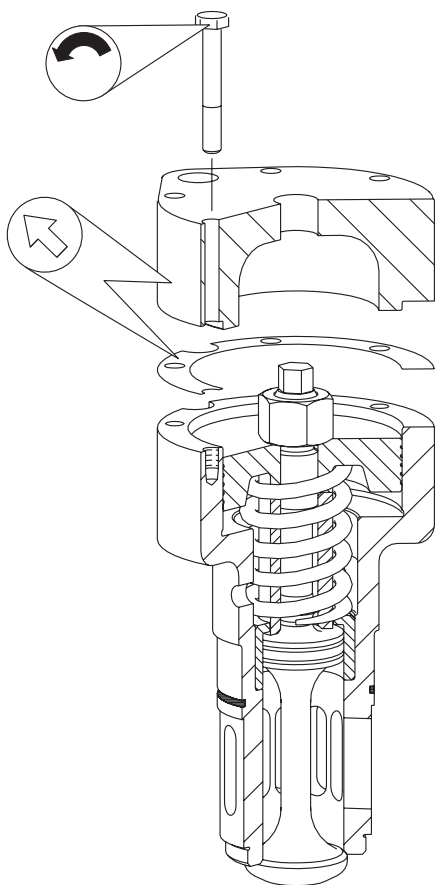
2.

D07-12



M90702-0219D02

1.



1. Remove the top cover fixing screws and remove the cover.

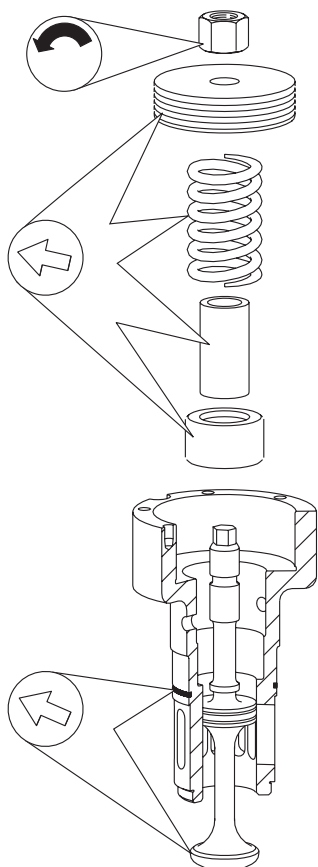
Remove and discard the gasket.

2. Screw off the valve spindle nut and remove the valve spindle from the valve housing.

Take out the piston, distance pipe, valve spring, and the liner.

Remove and discard the O-ring.

2.



M90702-0219001

M90702-0219002

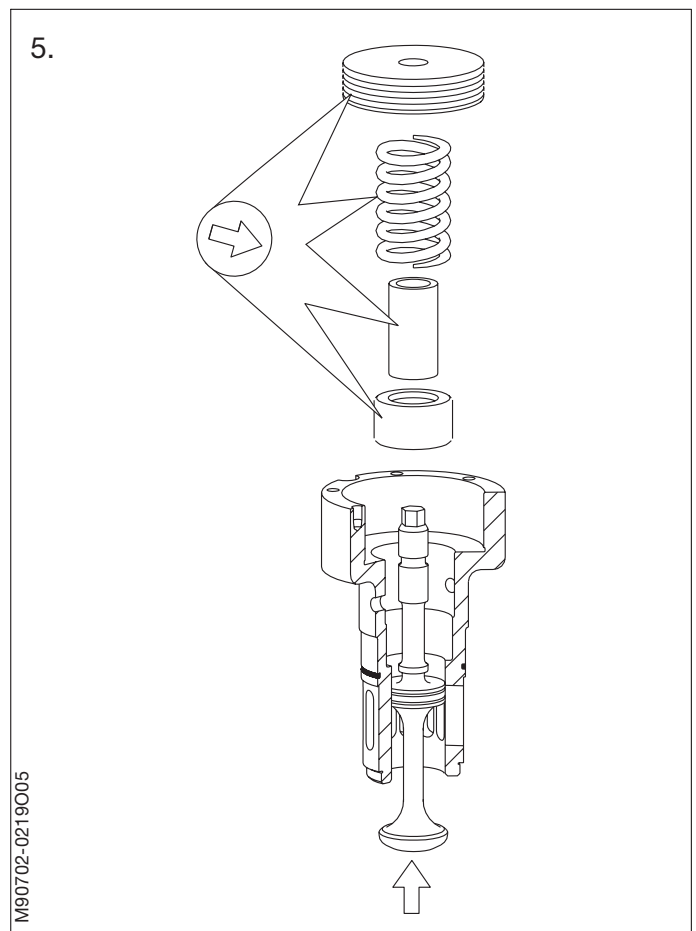
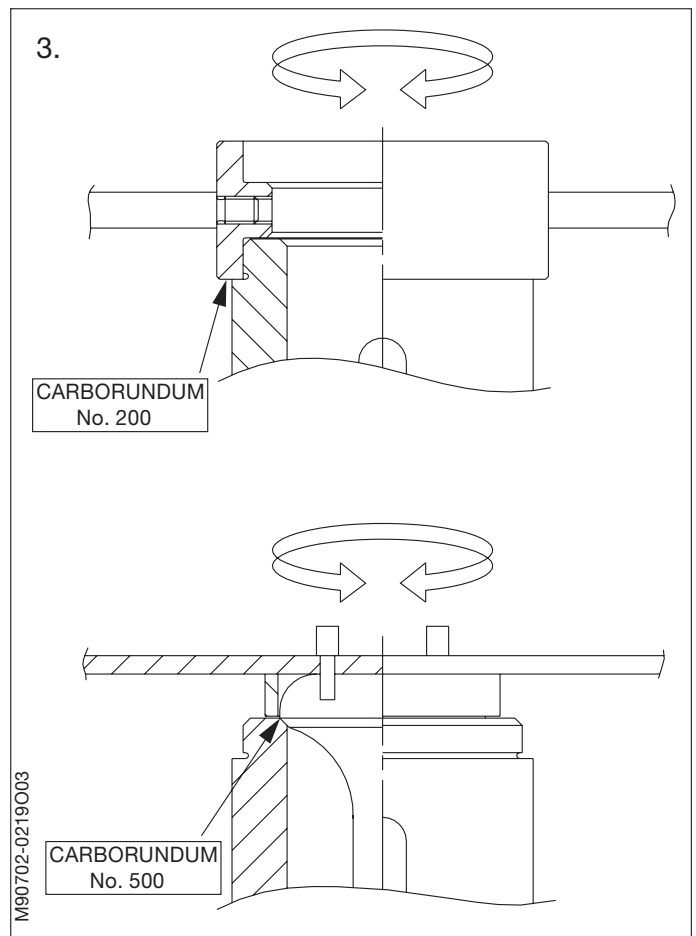
3. Grind the valve housing seating with the grinding ring after applying grinding paste. Carborundum No. 200.

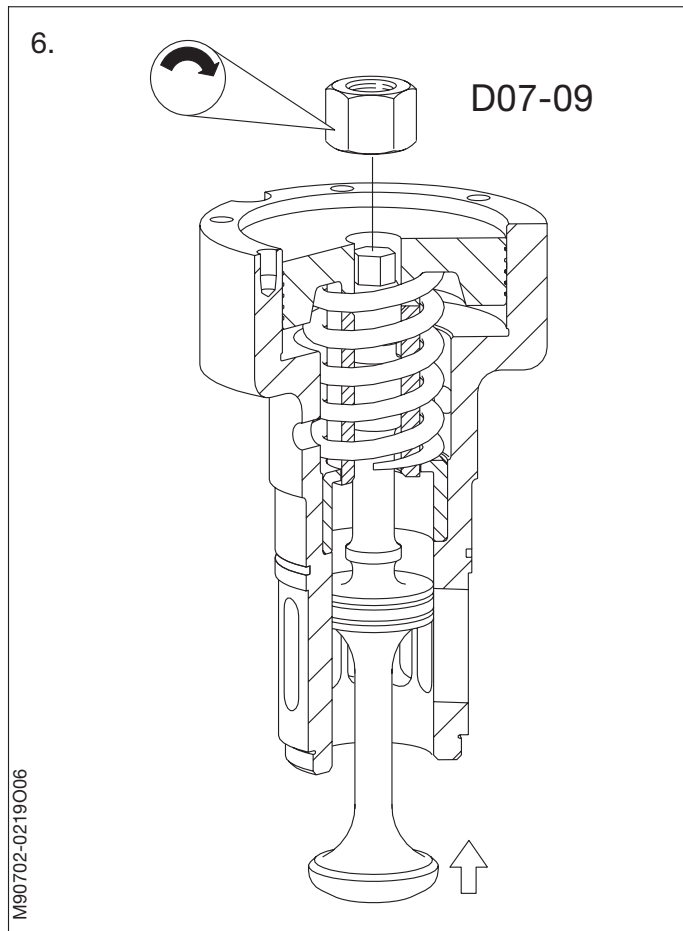
Grind the seatings of the spindle and housing to match, using the face wrench to rotate the valve spindle.
Carborundum No. 500.

4. Thoroughly clean all parts before reassembling the valve.

Lubricate all internal parts (sliding surfaces) with, for instance, Molybdenum Disulphide, MoS₂.

5. First mount the liner in the valve housing, then the valve spindle, the distance pipe, the spring and the piston.





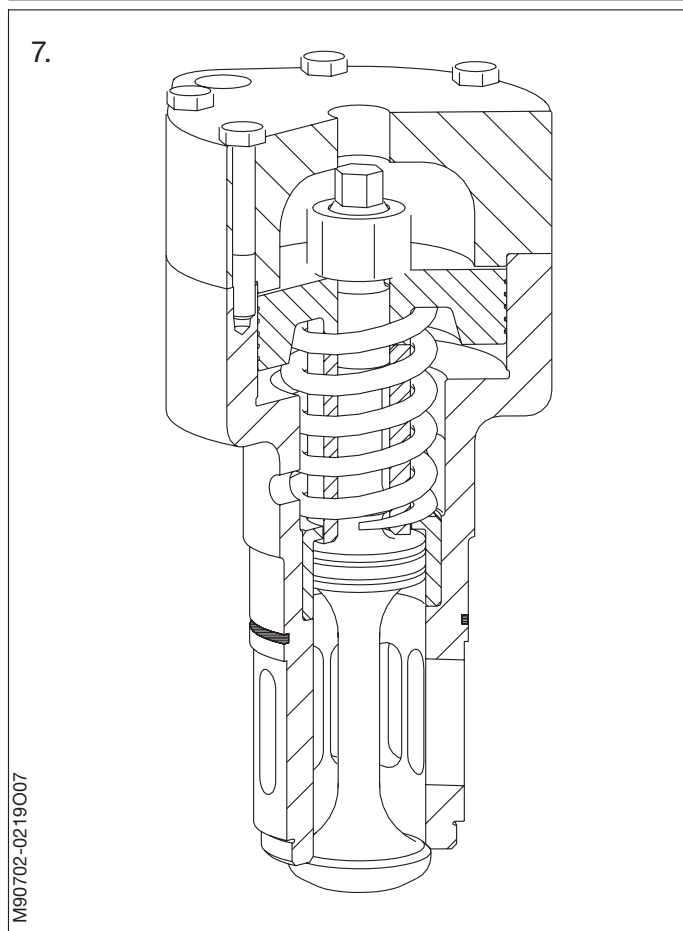
6. Mount the nut on the valve spindle and tighten the nut, see *D07-09*.

7. Place a new gasket on the valve housing.

Mount and tighten the top cover.

Replace the O-ring on the valve housing.

8. If the starting air valve is not to be mounted in the engine immediately after the overhaul, all openings of the valve should be covered with plastic to prevent dirt from entering the valve during storage.



1. Carefully clean the starting valve bore in the cylinder cover and, if necessary, recondition the seat for the starting valve in the bore.

See *Procedure 901-1.3*.

If not already done, replace the O-ring on the overhauled valve and lubricate with Neverseize or Molybdenum Disulphide, MoS₂.

For overhaul, see *Procedure 907-2.3*.

2. Mount the valve in the cylinder cover.
3. Mount the nuts and tighten-up either to specified torque or to specified torque and angle, **not** both.

See *Data D07-10 or D07-11*.

Mount the control air pipes and turn on starting air and control air.

